

Syllabus for Cloud Computing

CIS437(Section 01)

Fall 2026

Foundations of cloud computing: infrastructure/platform/software-as-a-service, virtualization, cloud platforms, on-demand/cloud applications. Challenges of cloud computing such as security and geographic implications, full-stack development, and big data. Cloud application development, deployment, and management is also covered.

Contact Information:

Instructor: Dr.Yong Zhuang
E-mail: yong.zhuang@gvsu.edu
Office: MAK D-2-234
Office Hours: T Th 4:00 pm - 5:00 pm, in-person (MAK D-2-234) and remote (Zoom)
Course Page: Blackboard & <https://gvsu-cis437.github.io/>
Zoom: Meeting ID: 396 668 6420, Password: 587684
Section 01: **Class time:** T Th, 2:30 pm - 3:45 pm, Synchronous
Room: Online
Midterm exam: Thursday, October 15, 2:30 pm - 3:45 pm
Final exam: Thursday, December 17, 2:00 pm - 3:50 pm

Course Objectives:

After completing this course, students should be able to:

- █ Define cloud computing and the role it plays in constructing modern software systems.
- █ Describe key issues/developments relevant to cloud computing such as security, geographic distribution, containerization, virtualization, privacy, etc.
- █ Develop cloud applications using modern cloud technologies.
- █ Compare Infrastructure-as-a-Service, Platform-as-a-Service, and Software-as-a-Service.
- █ Contrast the advantages and disadvantages of cloud applications

Prerequisites:

CIS337 or CIS457, admission to computing major, or permission of instructor

Course Materials:

Primary: Instructor's Lecture Notes, Open Educational Resources, and Handouts (via Blackboard)

Course Delivery - Online Synchronous:

This course will be delivered **in a synchronous format**

Expectations: I expect the following from you to ensure your success in this course:

- a) check [Blackboard](#) and [Course Website](#) on a regular basis for announcements, course material, sample code and assignments
- b) adhere to the CIS & GVSU policy of Academic Honesty

GVSU Blackboard: Course materials, assignments, grades, and announcements will be posted to Blackboard (<https://lms.gvsu.edu/>). It is your responsibility to stay informed.

Note: I use Blackboard's course announcements to communicate timely information about the course (e.g., changes to deadlines, etc). Ensure that your notification settings for Blackboard are set up to allow you to get updates as they are posted.

Grading Proportions:

Course Component	Overall Weight
In-class Activities / Quizzes	15%
Coursework	30%
Term Project	25%
Midterm Exam	15%
Final Exam	15%
Total	100%

The instructor reserves the right to make minor adjustments to the point distribution.

Grade A	Grade B	Grade C	Grades D & F
$A \geq 93\%$	$B+ \geq 87\%$	$C+ \geq 77\%$	$D+ \geq 67\%$
$A- \geq 90\%$	$B \geq 83\%$	$C \geq 73\%$	$D \geq 60\%$
	$B- \geq 80\%$	$C- \geq 70\%$	$F < 60\%$

Course Policies:

Assignments, Due Dates and Attendance:

 **Due dates:** All assignments will be due at 11:59 pm Michigan time on the due date.

 **Late policy:**

- ★ Work submitted after the due date will incur a 10% late penalty per day, with a minimum penalty of 10%.
- ★ **No assignment will be accepted more than 3 days late.**
- ★ In-class assignments are not accepted late regardless.

- ★ No assignment will be accepted late after the last day of class regardless of the number of days late. If you are struggling with meeting deadlines, please contact me as soon as possible!
- ★ All assignments, graded discussions, quizzes, exams, etc., are submitted electronically. No assignments are accepted via email, printed, or any other method.

Attendance:

- ★ Students are responsible for all material, announcements, and activities covered during class time. If you miss class, you must obtain lecture notes from a classmate.
- ★ This course uses brief in-class attendance quizzes to track student participation. These quizzes are not graded for correctness; they simply verify your attendance. You must submit the quiz during class time to receive credit.
- ★ Your attendance quiz grade contributes to 5% of your final course grade. Students who complete at least 80% of all attendance quizzes will receive full credit. If your attendance quiz submission rate is below 80%, you will receive zero for this portion.
- ★ There will be no make-up attendance quizzes unless prior arrangements are made or a documented excuse (e.g., illness, emergency) is provided.
- ★ University policy states: "In case of excessive absences, the instructor may refuse to grant credit for the course." [GVSU Attendance Policy](#).

 **Accommodations:** Inform your instructor of any required accommodations to ensure successful learning. Any impediments should be discussed with the instructor.

 The instructor reserves the right to modify course policies, calendars, and due dates.

 This course adheres to the GVSU policies available at www.gvsu.edu/coursepolicies/.

Academic Honesty:

All students are expected to adhere to the academic honesty standards of Grand Valley State University. In addition, students in this course are expected to adhere to the academic honesty guidelines set forth by the College of Computing. Details can be found at <https://www.gvsu.edu/computing/academic-honesty-93>

I believe in the value of learning from peers, both in class and within the broader community. Collaboration is encouraged, but it must be balanced with academic integrity. Here's how you can collaborate responsibly:

-  **Document Collaborations:** Clearly note any collaborations on individual assignments.
-  **No Code Sharing:** Direct electronic transfer of code between students is not allowed.
-  **Cite Internet Sources:** If you use code from the internet, provide an active link and ensure it doesn't constitute the entire solution.
-  **Engage Online Respectfully:** Participate in forums for discussion, not for sharing solutions or soliciting complete answers.
-  **Discuss Conceptually:** Talk about problems using non-technical, conceptual language rather than sharing specific code.
-  **ChatGPT/LLM statement:** I do not mind if you use ChatGPT to help you work through problems, however do not simply copy and paste its output into your assignment. This is the same as plagiarism/cheating as you are presenting work that is **not your own**.
-  Ultimately, you are responsible for all aspects of your submissions. You should be able to explain and defend your submission if the work is entirely your own.

Academic Resources: GVSU also provides opportunities for students to improve their academic skills through resources such as:

- **The writing center:** The Fred Meijer Center for Writing, with locations at the Allendale and Pew/Downtown Grand Rapids campuses, can assist you with writing for any of your classes. For more information about these services and locations, please visit their website: <http://www.gvsu.edu/wc/>
- **Speech lab:** The Grand Valley Speech Lab is a peer-to-peer communication center that helps students with all elements of oral presentations. For more information about this service, please visit their website: <https://www.gvsu.edu/speechlab/>
- **Research consultants:** The Center for Scholarly and Creative Excellence (CSCE) promotes a culture of active, engaged, ethical scholarship. It supports innovative faculty and student research and collaborative partnerships in the broader community. For more information, please visit their website: <https://www.gvsu.edu/csce/>
- **Library:** GVSU's library offers a vast collection of online resources available for students. Visit their website for more details: <https://www.gvsu.edu/library/>
- **Disability support resources:** If any student in this class has special needs because of a disability, please contact Disability Support Resources at <http://www.gvsu.edu/dsr/> (DSR) at 616-331-2490.

Religious Observance: The university recognizes and respects religious traditions. If you require special accommodations for religious observances, inform the instructor in advance.

Emergency Procedures:

- 🔔 **In Case of Emergency Fire:** Immediately proceed to the nearest exit during a fire alarm. Do not use elevators. More information is available on the University's Emergency website located at <http://www.gvsu.edu/emergency>.

Tentative Course Content (subject to change throughout the semester):

🔪 **September 6–7, Labor Day Recess:** No classes!

🔪 **October 26–27, Fall Break:** No classes!

🔪 **November 25–29, Thanksgiving Recess:** No classes!

Module	Topics Covered
1	Syllabus, Introduction
2	Cloud Platforms
3	Cloud Shell, Cloud Applications
4	* as a Service, Microservices, Serverless Computing
5	Security and Geographic Implications
6	Various Sizes of Data
7	APIs
8	Cloud Run, Cloud Build
9	Cloud Operations
10	Machine Learning and Generative AI
11	Midterm Exam (October 15, 2:30 PM - 3:45 PM)
12	Final Exam (December 17, 2:00 PM–3:50 PM)
